ISLP Chapter 3: Linear Regression Study Guide

A Visual, Mathematical, and Code-First Terrain Map

Linear Regression is the fundamental parametric model of supervised learning. It maps a linear relationship between features and continuous targets. This study guide breaks Chapter 3 down into five core pillars, highlighting the underlying linear algebra, statistical inference, diagnostic tests, and parametric tradeoffs.

1. Coefficient Estimation & Least Squares

The simple linear regression model is written as: $Y = \beta_0 + \beta_1 X + \varepsilon$. We fit this model by minimizing the Residual Sum of Squares: $\mathrm{RSS} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$. In multiple linear regression, the closed-form Ordinary Least Squares (OLS) solution is the Normal Equation:

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

Technical Bottleneck: Inverting the Gram matrix X■X requires roughly $O(p^3)$ operations where p is the number of features. When features are perfectly collinear, the matrix is singular and cannot be inverted, breaking OLS.

2. Statistical Inference & Hypothesis Testing

To test whether features have a statistically significant relationship with the response, we perform t-tests on individual coefficients. However, when examining multiple predictors simultaneously, the multiple testing problem causes individual t-tests to falsely flag significance (high family-wise error rate). To verify the global model's significance, we must look out for the global F-statistic:

$$F = \frac{(\mathrm{TSS} - \mathrm{RSS})/p}{\mathrm{RSS}/(n - p - 1)}$$

Confidence vs. Prediction Intervals: When making predictions, confidence intervals measure the uncertainty in the average fit. Prediction intervals measure the uncertainty in a single new observation. Because a single observation contains the irreducible error variance (epsilon), the prediction interval is always much wider than the confidence interval (see Figure 1).

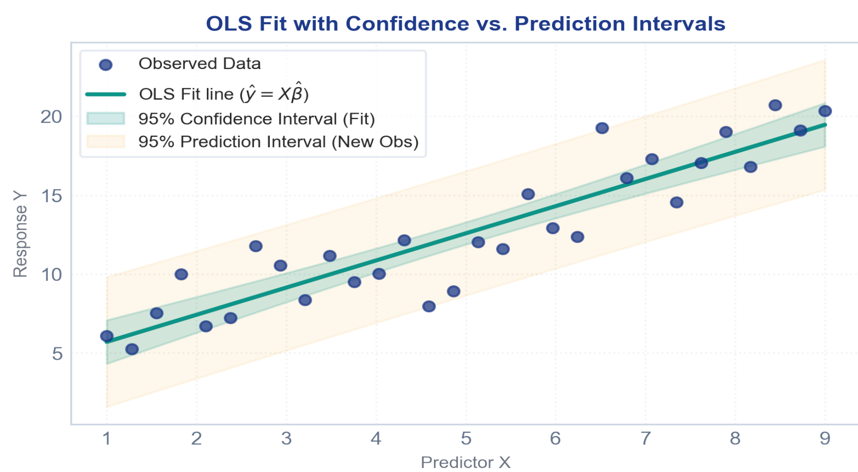

Figure 1: OLS regression line fitted with 95% Confidence Interval vs. 95% Prediction Interval bounds.

3. Assumptions, Diagnostics, and Pitfalls

Linear regression relies on several crucial assumptions. Diagnostic plots are used to identify when these assumptions are violated, as well as finding influential data points that degrade the model fit:

• **Non-Linearity:** Identified by plotting residuals vs. fitted values. A distinct non-linear shape points to a non-linear relationship (see Figure 2).
• **Heteroscedasticity:** Non-constant variance of error terms. Visible as a funnel-shaped pattern in the residuals plot.
• **Collinearity:** High correlation between predictors. Quantified using the **Variance Inflation Factor (VIF)**: $\mathrm{VIF}(\hat{\beta}_j) = \frac{1}{1 - R^2_{X_j|X_{-j}}}$, which measures how much the variance of a coefficient is inflated.
• **Outliers & Leverage:** Outliers are points where the true target is far from the model's prediction. **High-leverage** points are observations with unusual predictor values, measured by leverage: $h_i = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_{k=1}^{n} (x_k - \bar{x})^2}$. Influential points (high residual + high leverage) distort the fit.

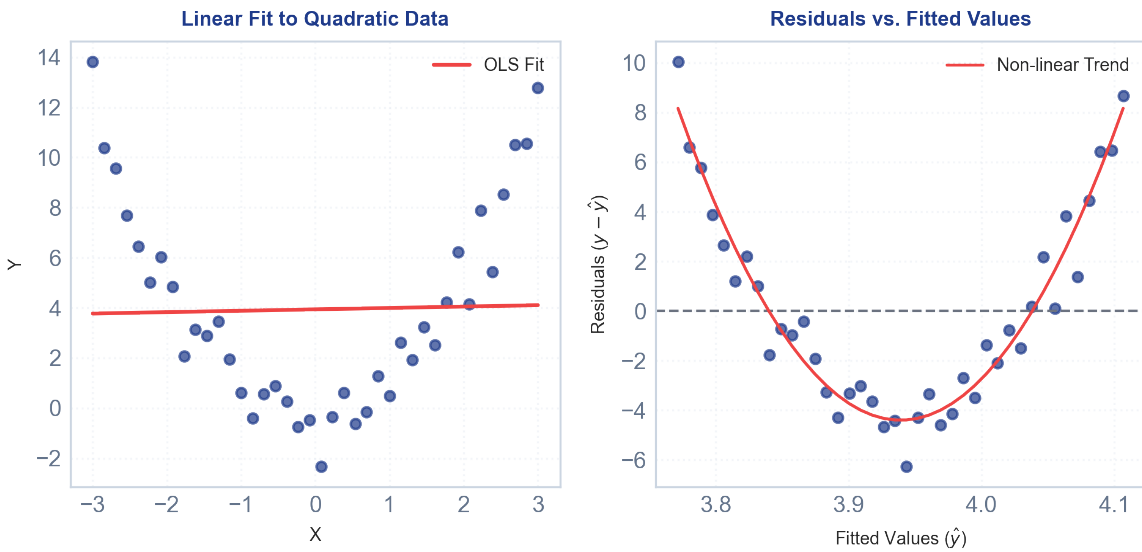

Figure 2: Left: Poor linear fit to curved data. Right: Residuals vs. fitted plot showing a clear curved pattern, indicating non-linearity.

4. Extending the Linear Model (Interactions & Non-linearity)

Standard OLS assumes additivity (features affect the response independently) and linearity. We extend the model via:
• **Interaction Terms:** Adding multiplicative terms to model joint synergy: $Y = \beta_0 + (\beta_1 + \beta_3 X_2) X_1 + \beta_2 X_2 + \varepsilon$. This means the effect of X■ changes dynamically as X■ changes.
• **Polynomial Features:** Adding powers of predictors (e.g. X^2) to capture curves while keeping the model linear in coefficients.

Hierarchical Principle: If an interaction term X■X■ is included in a regression model, the main effects X■ and X■ must be included in the model, even if their individual p-values are not statistically significant.

5. Parametric vs. Non-Parametric (Linear Regression vs. KNN)

Linear regression is a **parametric** model (it assumes a rigid linear shape), while K-Nearest Neighbors (KNN) is **non-parametric** (makes no shape assumptions). Non-parametric methods perform well when the true relationship is highly non-linear, but they suffer severely from the **Curse of Dimensionality**. In high-dimensional spaces (large p), points are spread extremely far apart, causing KNN's performance to collapse, meaning OLS is often superior.

From-Scratch Multiple Linear Regression (NumPy)

Here is the standard matrix implementation of Multiple Linear Regression, performing fit (Normal Equation) and inference predictions:

```
import numpy as np

class MultipleLinearRegression:
    def __init__(self, fit_intercept=True):
        self.fit_intercept = fit_intercept
        self.beta = None

    def fit(self, X, y):
        # 1. Add column of ones for intercept (bias) if required
        if self.fit_intercept:
            ones = np.ones((X.shape[0], 1))
            X = np.hstack([ones, X])

        # 2. Solve normal equation: beta = (X^T * X)^(-1) * X^T * y
        XtX = X.T @ X
        self.beta = np.linalg.inv(XtX) @ X.T @ y
        return self

    def predict(self, X):
        if self.fit_intercept:
            ones = np.ones((X.shape[0], 1))
            X = np.hstack([ones, X])
        return X @ self.beta
```

ML INSIGHT: *Collinearity does not affect prediction performance or R^2 directly, but it inflates coefficient variances and makes the model interpretation unreliable. Check VIF scores if you are doing inference and need to identify true feature importances.*

Linear Regression Quick Reference Sheet

Key formulas, diagnostic tests, and concepts in ISLP Chapter 3.

Concept / Formula Name	Mathematical Notation	ML Context & Practical Application
Linear Model	$Y = \beta_0 + \beta_1 X + \varepsilon$	Parametric linear assumption relating predictors to output.
Normal Equation (OLS)	$\hat{\beta} = (X^T X)^{-1} X^T y$	Analytical solution to coefficient estimation. Inverse complexity $O(p^3)$.
Residual Sum of Squares	$\mathrm{RSS} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$	Objective function minimized by Ordinary Least Squares.
Global F-Statistic	$F = \frac{(\mathrm{TSS} - \mathrm{RSS})/p}{\mathrm{RSS}/(n-p-1)}$	Tests joint significance of all features, avoiding multiple testing error.
Variance Inflation Factor	$\mathrm{VIF}(\hat{\beta}_j) = \frac{1}{1 - R^2_{X_j \mid X_{-j}}}$	Quantifies collinearity. VIF > 5 or 10 indicates high multicollinearity.
Leverage Statistic	$h_i = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_{k=1}^{n} (x_k - \bar{x})^2}$	Measures predictor outlier severity. Outlier points skew the OLS fit.
Interaction Model	$Y = \beta_0 + (\beta_1 + \beta_3 X_2) X_1 + \beta_2 X_2 + \varepsilon$	Models synergy between predictors. Governed by Hierarchical Principle.
R^2 Statistic	$R^2 = 1 - (\mathrm{RSS} / \mathrm{TSS})$	Proportion of variance explained by model. Stretches [0, 1].
Residual Standard Error	$\mathrm{RSE} = \mathrm{sqrt}(\mathrm{RSS} / (n - p - 1))$	Estimator of standard deviation of error term (irreducible error).